

Project Summary Report: Security Company Network

Prepared by:

Jana Mufti – S21106991

Afnan Kamel - S22208066

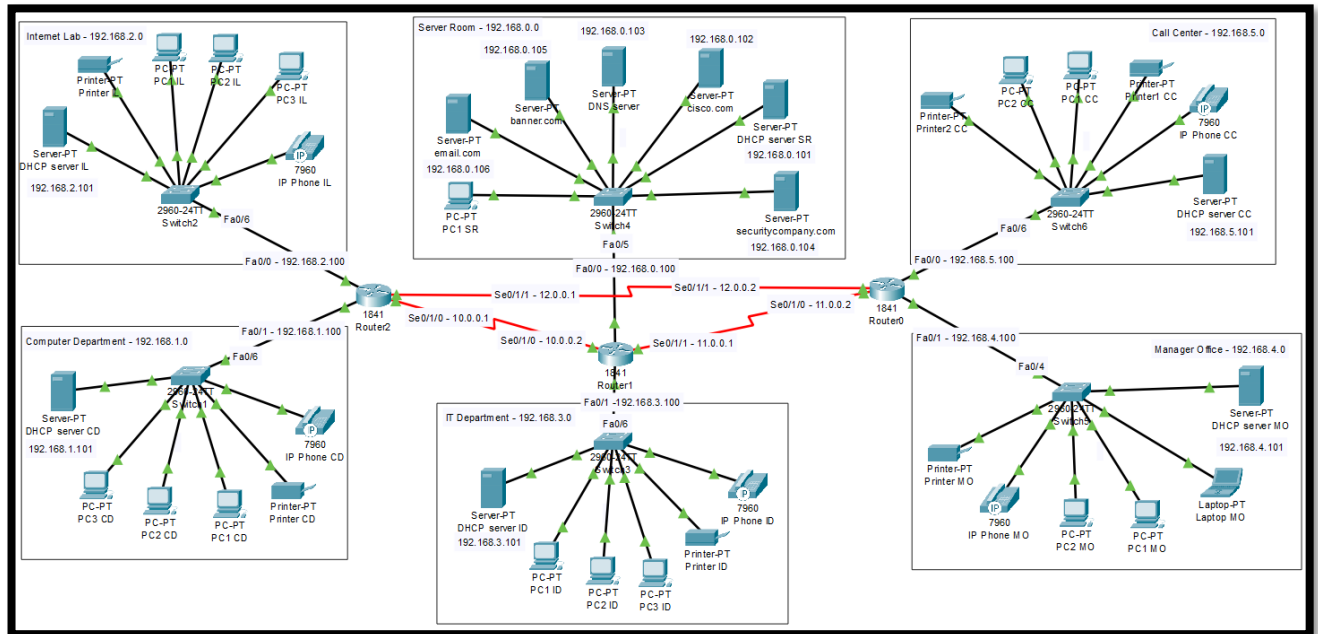
Aya Mohammed - S21207588

Afraha Bashaddadah - S22107697

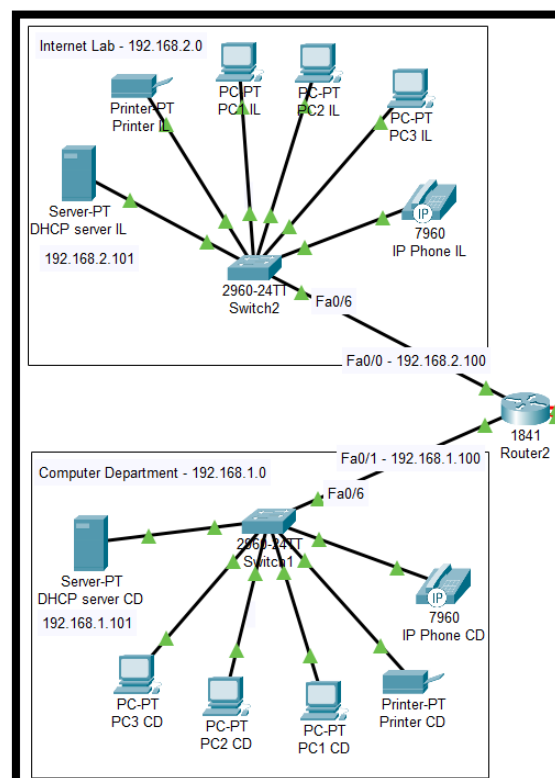
Supervised by: Dr. Mohammad Nauman

Final Project – Spring 2024

- The security company network consists of 6 LANs and 3 routers that connect them. Each pair of LANs is using one router.

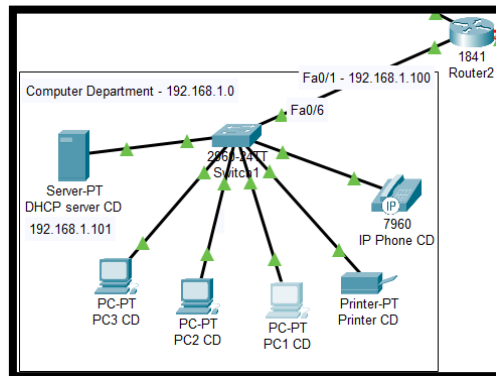


- These are the first two LANs, which are the internet lab of the company and the computer department. Each consists LAN has the capacity of 50 end devices, and each one has a DHCP server to provide IP addresses for all the end devices.

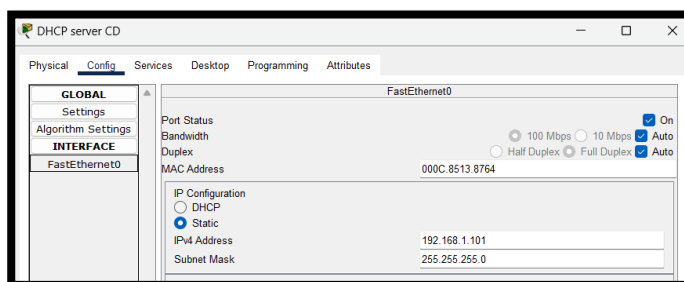
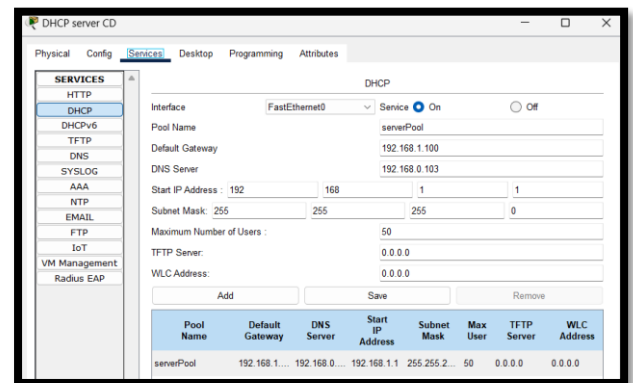
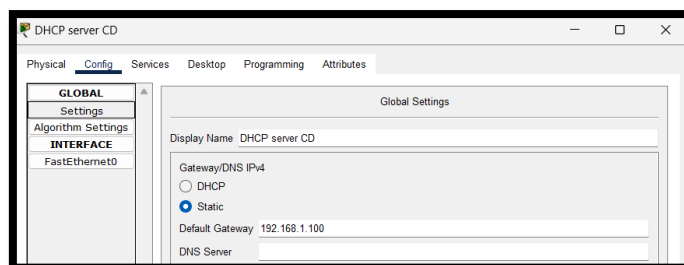


Final Project – Spring 2024

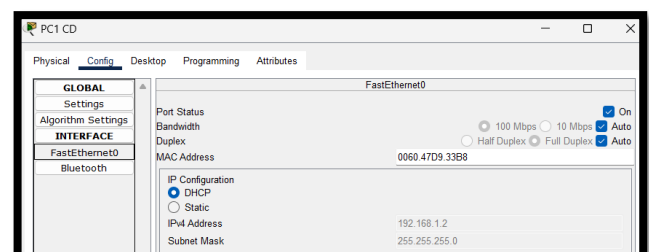
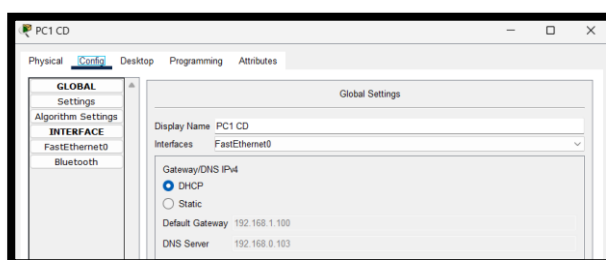
- The first LAN is the computer department with an IP network address of 192.168.1.0. It consists of 3 PCs, 1 printer, and 1 phone, all of which are end devices, along with 1 DHCP server.



- This is the configuration of the DHCP server for the computer department.

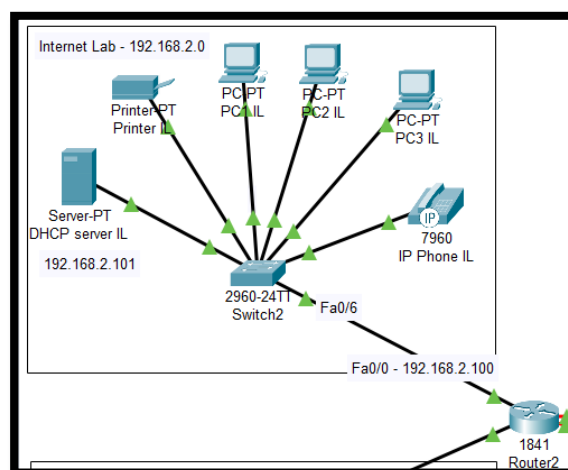


- The configuration of one of the end devices in the computer department.

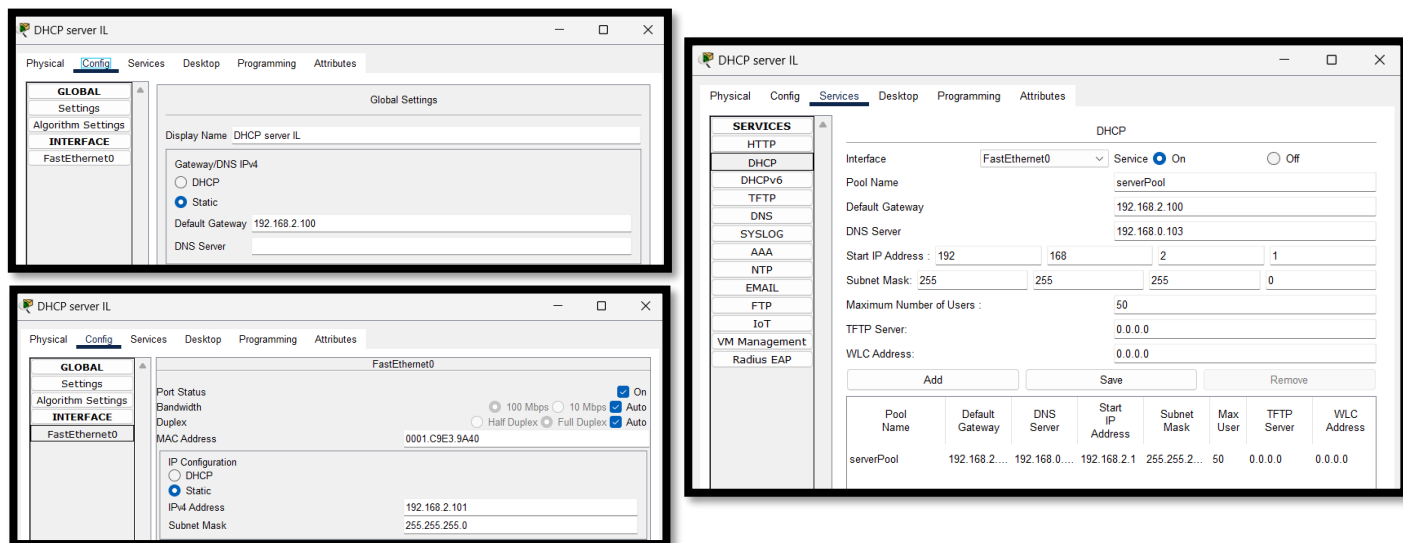


Final Project – Spring 2024

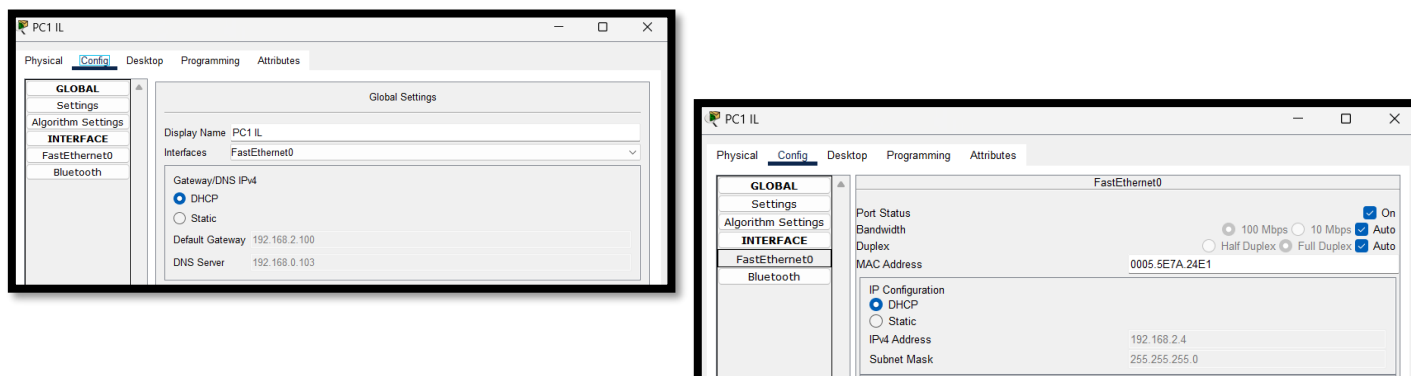
- The second LAN is the internet lab with an IP network address of 192.168.2.0. It consists of 3 PCs, 1 printer, and 1 phone, all of which are end devices, along with 1 DHCP server.



- This is the configuration of the DHCP server for the internet lab.

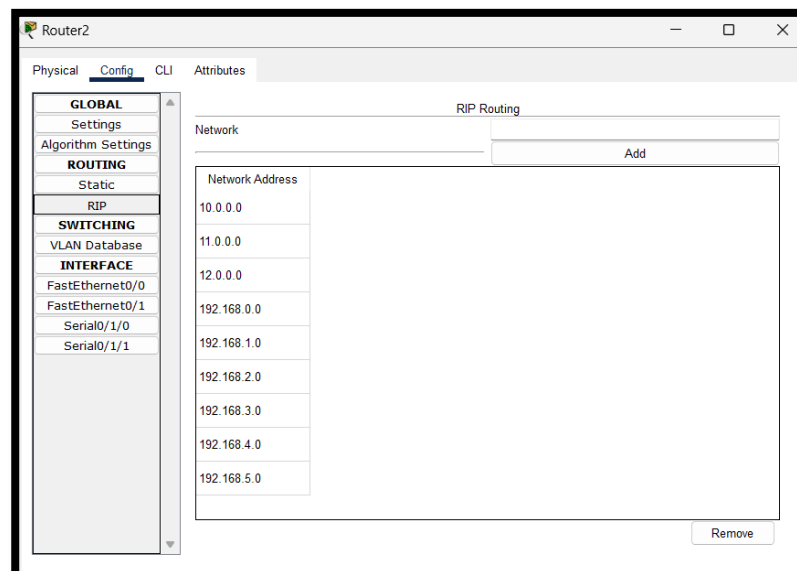
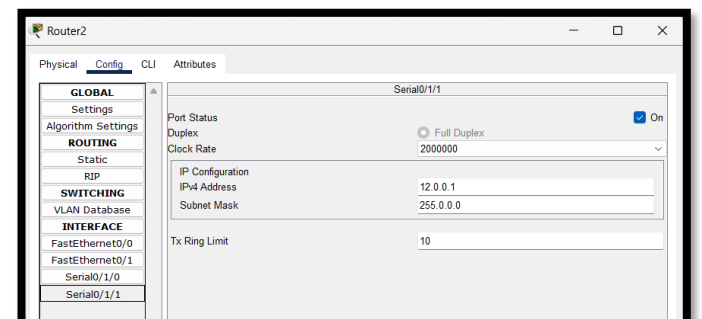
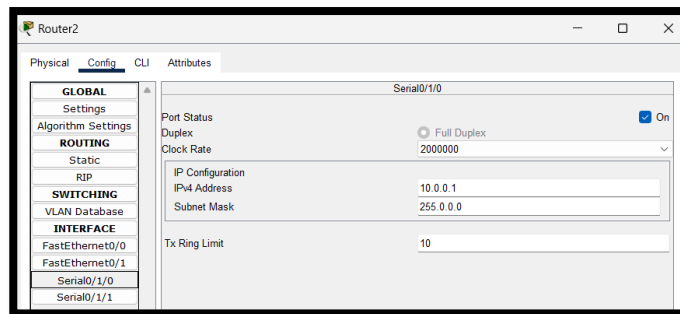
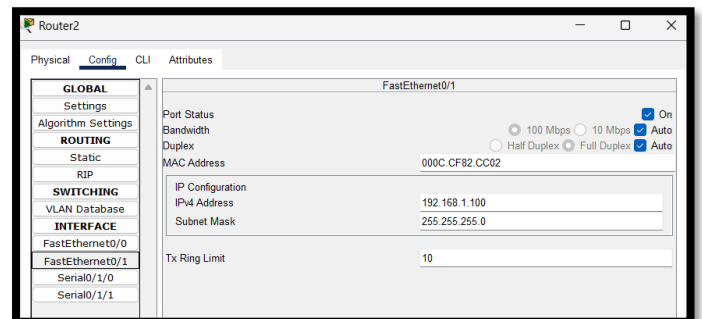
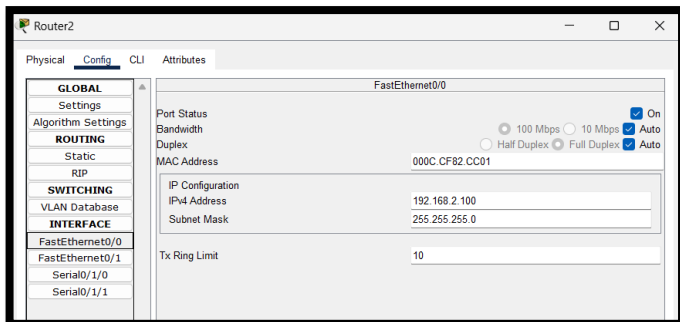


- The configuration of one of the end devices in the internet lab.



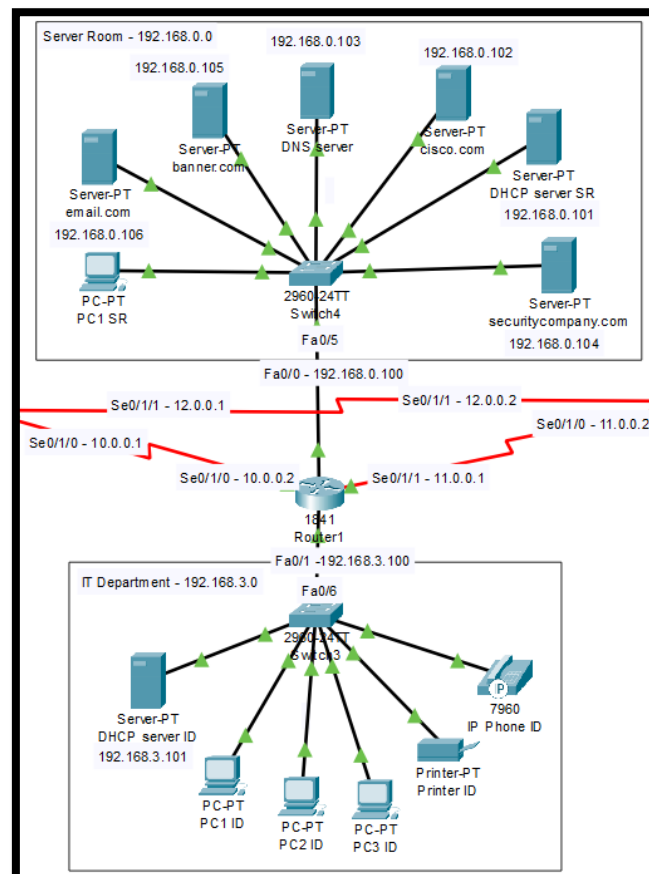
Final Project – Spring 2024

- This is the configuration of the first router, "Router2," which is connected to two LANs (computer department and internet lab) and is also connected to the other routers. To connect it to multiple routers, we inserted WIC-2T modules into the physical device. It is using RIP routing.

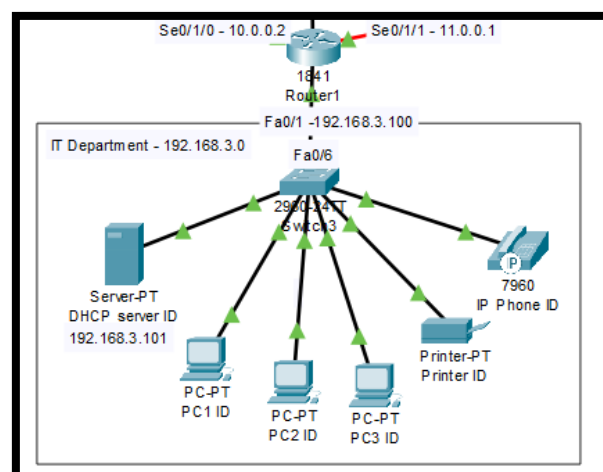


Final Project – Spring 2024

- These are the other two LANs, which are the server room and IT department of the company. Each consists LAN has the capacity of 50 end devices, and each one has a DHCP server to provide IP addresses for all the end devices.

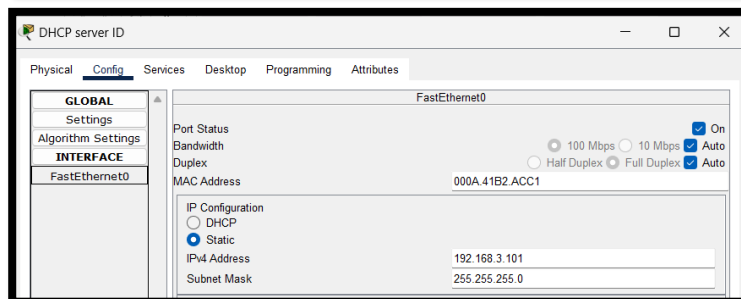
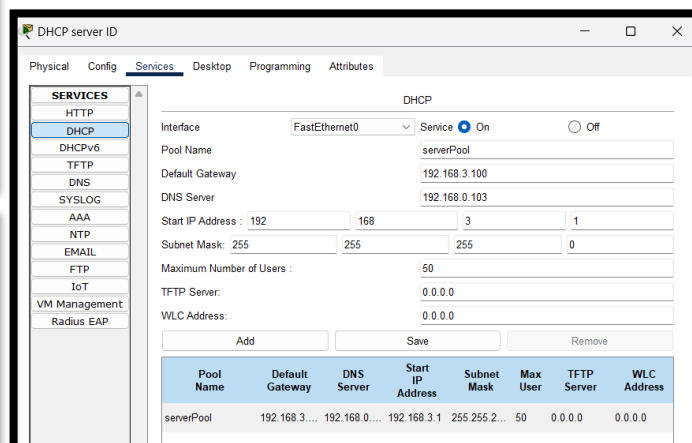
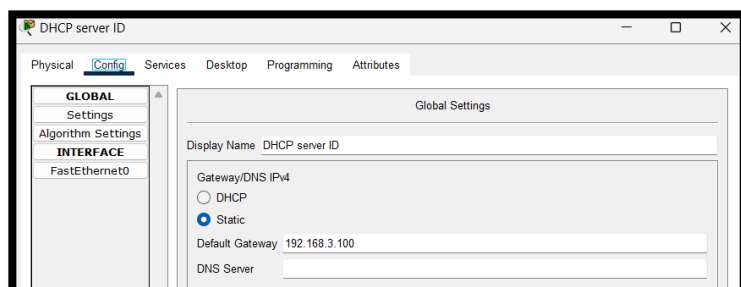


- The third LAN is the IT department with an IP network address of 192.168.3.0. It consists of 3 PCs, 1 printer, and 1 phone, all of which are end devices, along with 1 DHCP server.

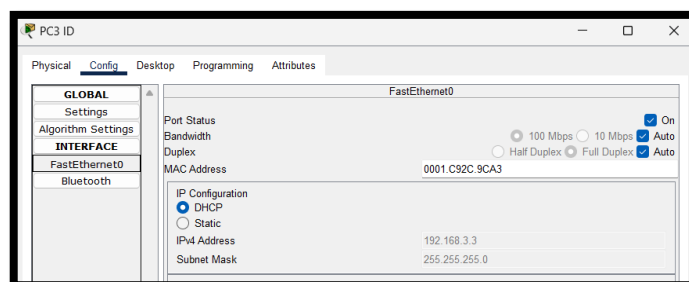
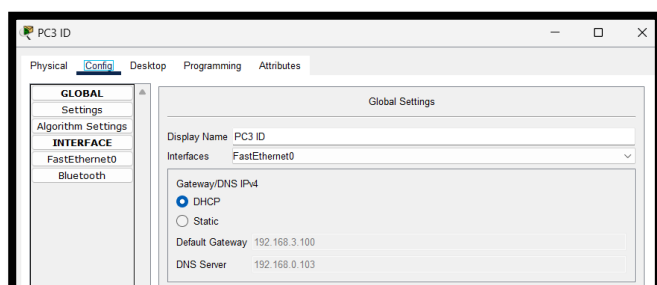


Final Project – Spring 2024

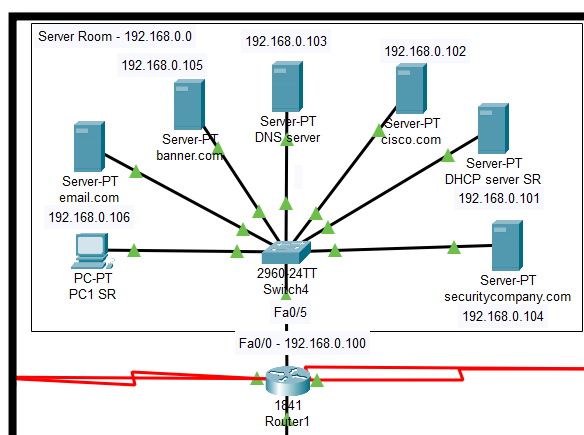
- i. This is the configuration of the DHCP server for the IT department.



- ii. The configuration of one of the end devices in the IT department.

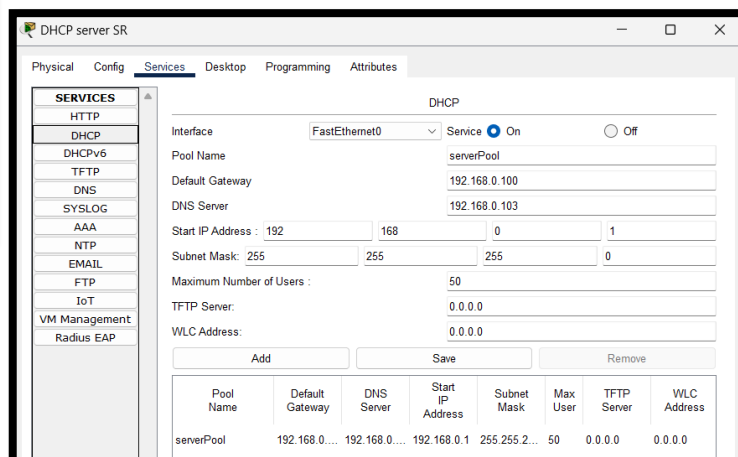
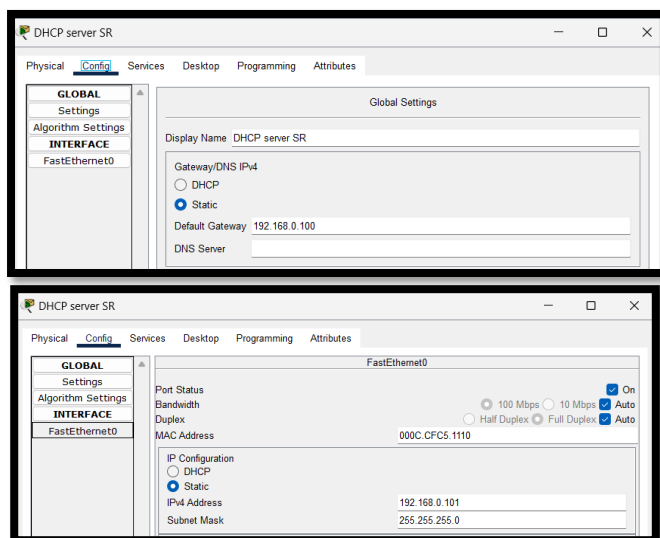


- The fourth LAN is the server room with an IP network address of 192.168.0.0. It consists of 1 PC (end device), 1 DNS server which translates human-readable domain names into machine-readable IP addresses, 4 web servers (email.com, banner.com, securitycompany.com, and cisco.com), and 1 DHCP server.

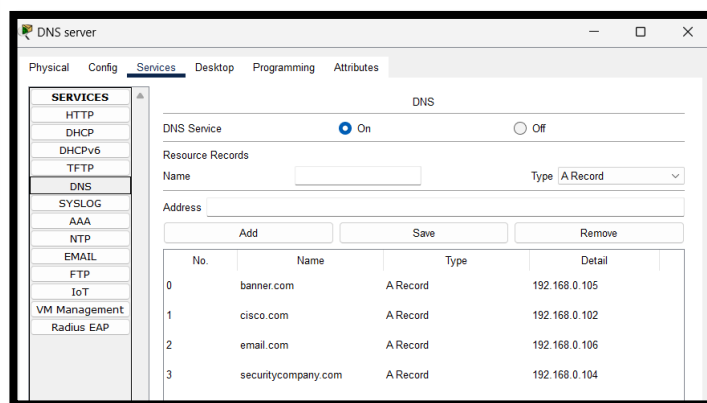
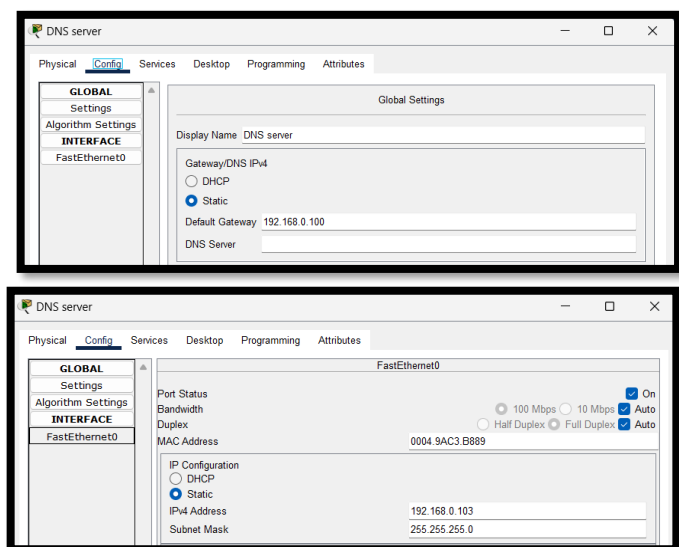


Final Project – Spring 2024

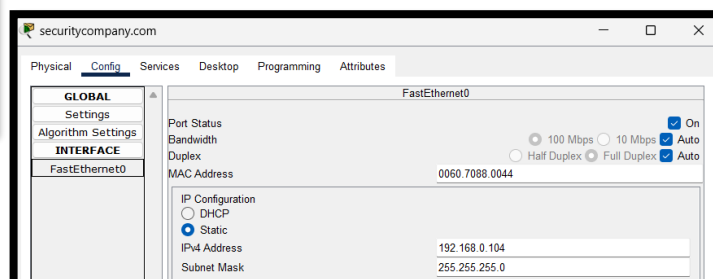
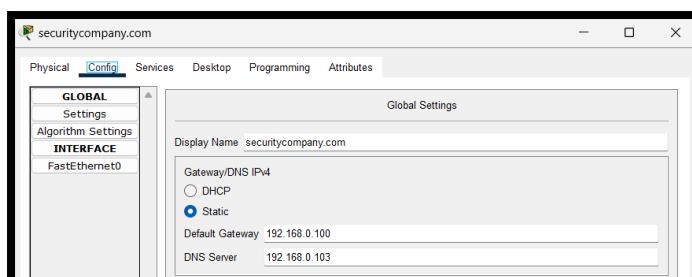
i. This is the configuration of the DHCP server for the server room.



ii. This is the configuration of the DNS server for the server room.

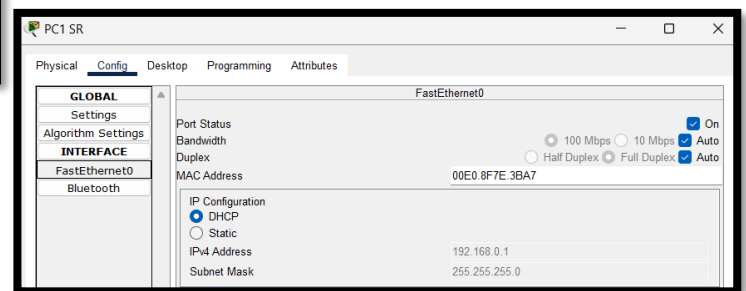
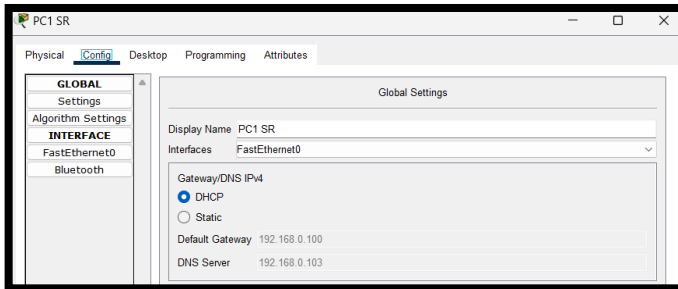


iii. The configuration of one of the web servers in the server room.

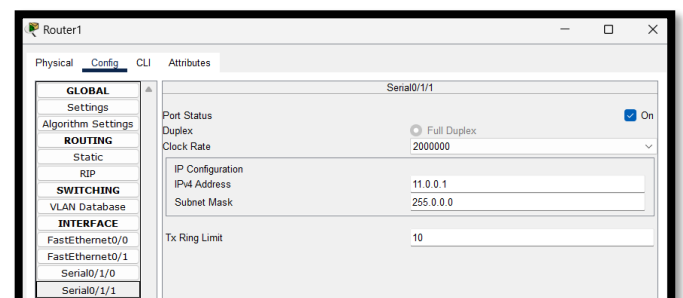
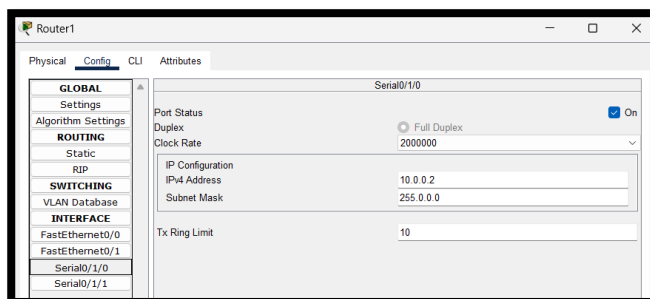
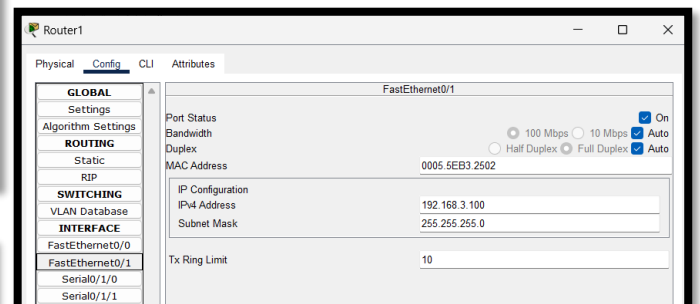
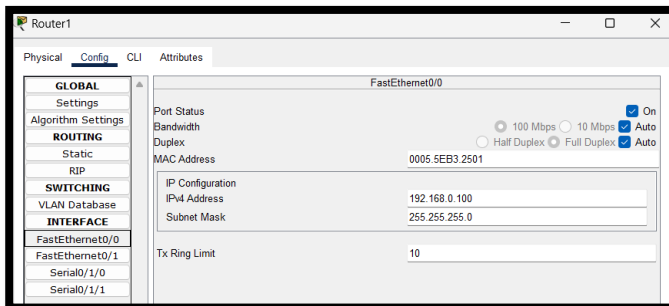


Final Project – Spring 2024

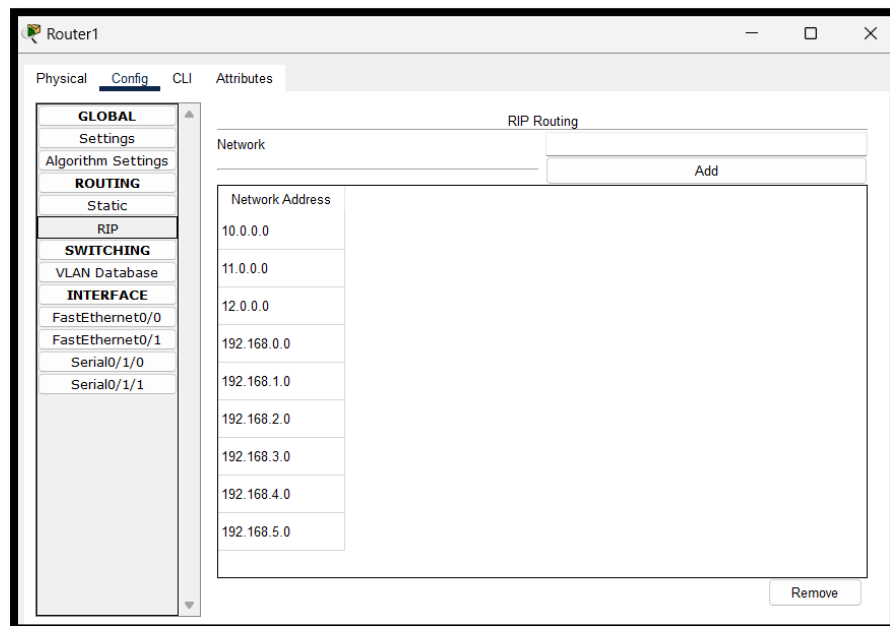
- iv. The configuration of one of the end devices in the server room.



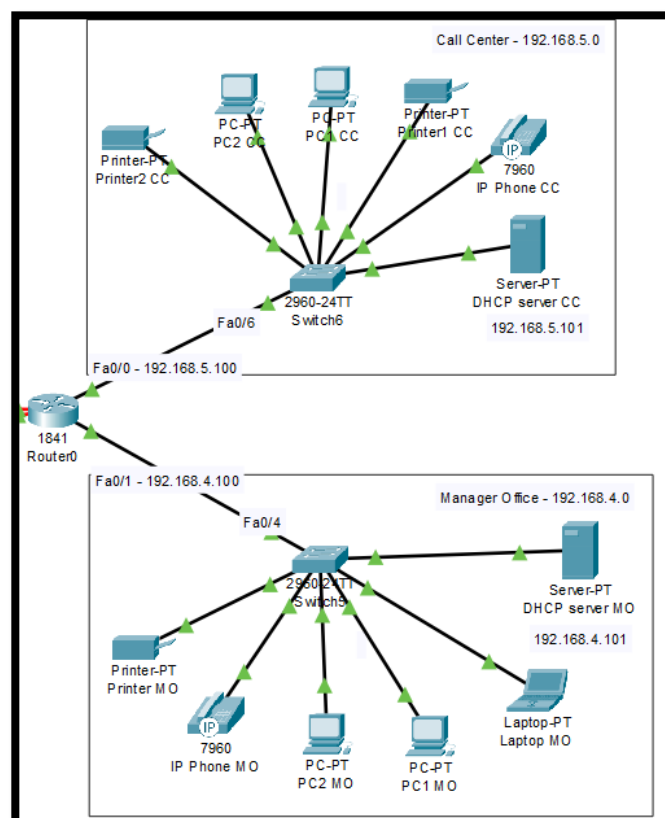
- This is the configuration of the first router, "Router1," which is connected to two LANs (IT department and server room) and is also connected to the other routers. To connect it to multiple routers, we inserted WIC-2T modules into the physical device. It is using RIP routing.



Final Project – Spring 2024

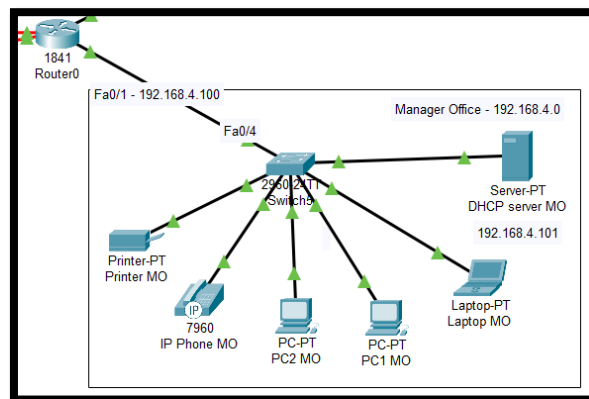


- These are the other two LANs, which are the call center and manager office of the company. Each consists LAN has the capacity of 50 end devices, and each one has a DHCP server to provide IP addresses for all the end devices.

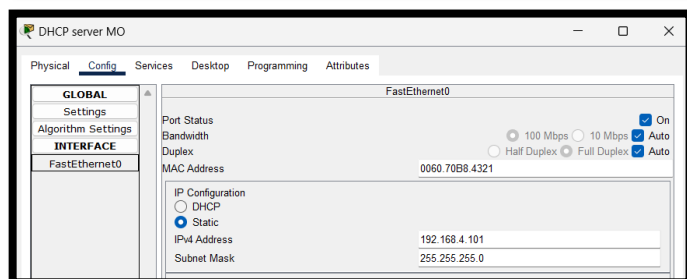
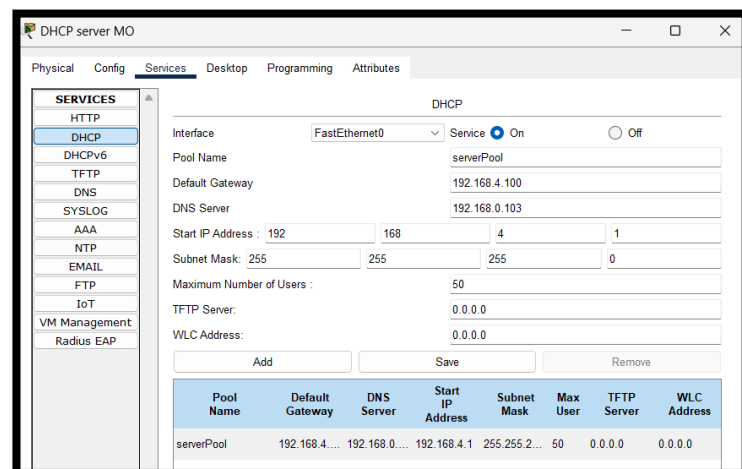
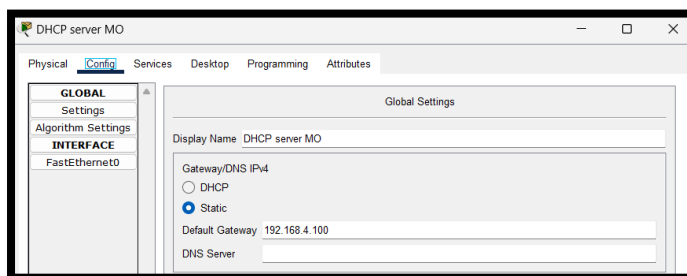


Final Project – Spring 2024

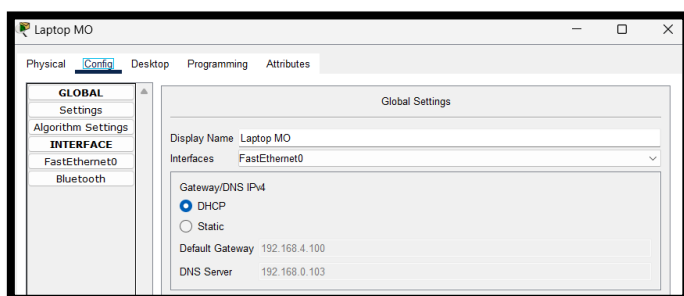
- The fifth LAN is the manager office with an IP network address of 192.168.4.0. It consists of 2 PCs, 1 laptop, 1 printer, and 1 phone, all of which are end devices, along with 1 DHCP server.



- i. This is the configuration of the DHCP server for the Manager office.

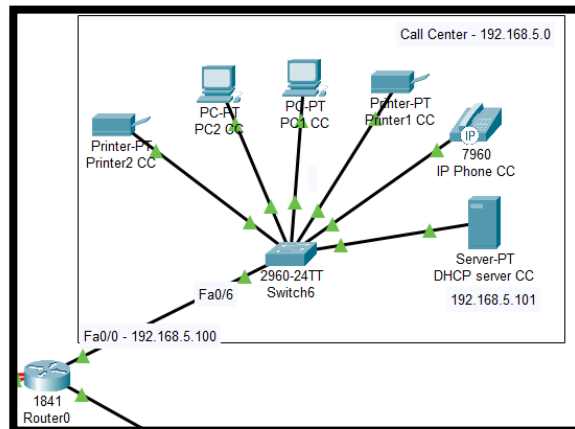


- ii. The configuration of one of the end devices in the Manager office.

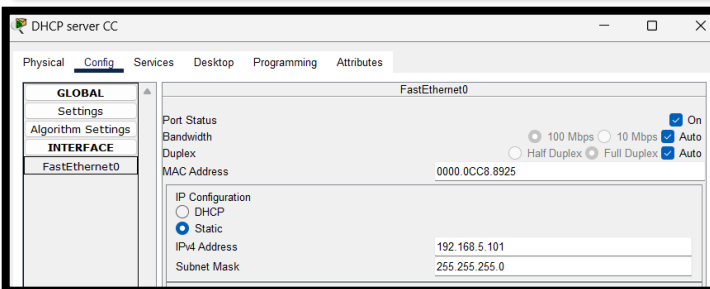
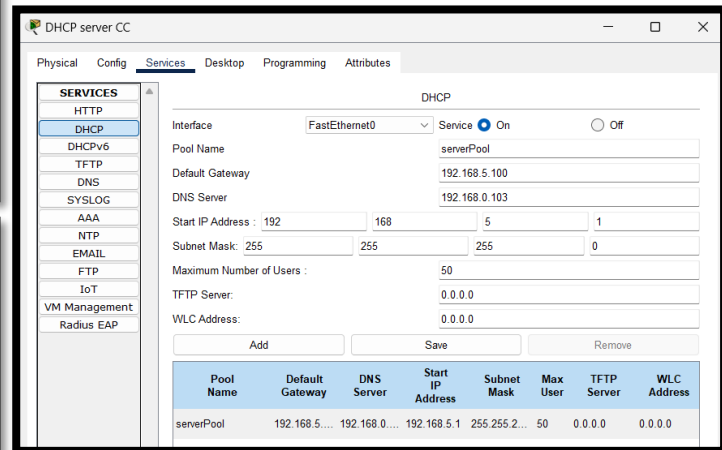
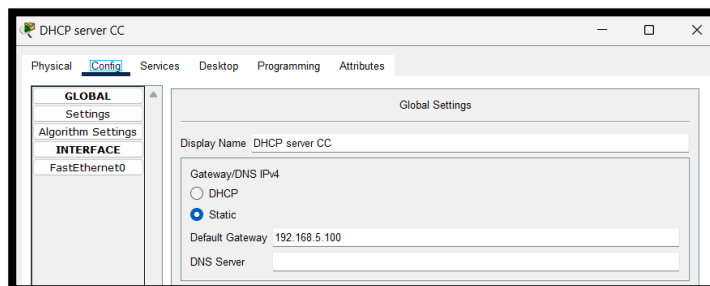


Final Project – Spring 2024

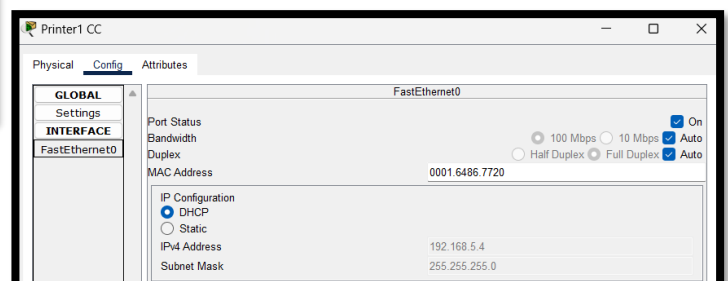
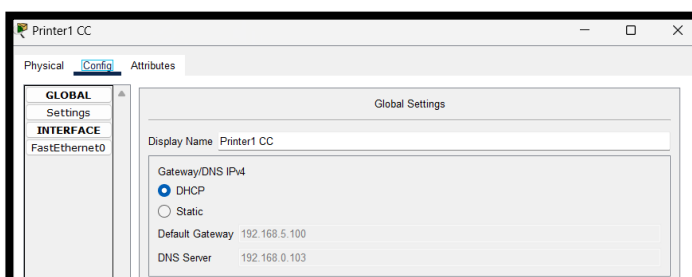
- The sixth LAN is the call center with an IP network address of 192.168.5.0. It consists of 2 PCs, 2 printers, and 1 phone, all of which are end devices, along with 1 DHCP server.



- i. This is the configuration of the DHCP server for the call center.



- ii. The configuration of one of the end devices in the call center.



Final Project – Spring 2024

- This is the configuration of the first router, "Router0," which is connected to two LANs (call center and manager office) and is also connected to the other routers. To connect it to multiple routers, we inserted WIC-2T modules into the physical device. It is using RIP routing.

